2025 IEEE International Workshop on Metrology for Sustainability (MetroSustainability) | 979-8-3315-8566-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/MetroSustainability67617.2025.11548174

Unlocking the Potential of Cloud Computing for Sustainable Energy and Cost-Efficient Artificial Intelligence Workloads

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***Abstract*—The escalating energy consumption and static resource constraints of large-scale Artificial Intelligence (AI) workloads present a critical challenge to sustainable computing. This paper introduces a highly scalable serverless parallel processing architecture designed to overcome the limitations of scalability and resource inefficiency. The novel technique employs a distributed map state within a generalized workflow orchestration service to handle massive data concurrency, validated on datasets exceeding 50,000 files. Resource efficiency is optimized through a data-driven function power tuning methodology that automatically balances cost and performance, preventing resource over-provisioning. The system integrates robust environmental accountability via a customer carbon footprint tool to provide granular, verifiable Greenhouse Gas Protocol-compliant emissions data. Empirical results confirm the architectural superiority, achieving a consistent ∼6x speedup in execution time and ∼80 percent reductions in compute, cost, and CO_2 emissions. Furthermore, task-specific model fine-tuning significantly improves predictive accuracy, making this architecture platform-agnostic, cost-effective, and environmentally responsible pathway for deploying AI in diverse, compute-intensive domains.**

***Index Terms*—artificial intelligence (AI) workloads; serverless computing; parallel processing; resource optimization; sustainable AI; cloud carbon accounting; distributed map; model fine-tuning.**

I. Introduction

The escalating energy consumption and environmental impact of artificial intelligence (AI) workloads [1], particularly from large language models (LLMs) like BERT and GPT, present a significant challenge to sustainable computing. The sheer scale of modern AI presents profound resource challenges. By 2030, AI data center clusters are estimated to consume over 1000 terawatt-hours annually which is Japan's annual electricity consumption. These hyperscale data centers, often demanding upwards of 100 MW—energy consumption comparable to a small city—require water-intensive cooling systems, which further increase power needs. Consequently, these massive facilities contribute a substantial portion of global greenhouse gas emissions [2]. For instance, training a single large model like DistilBERT requires significant energy and computational resources. Beyond environmental damage, this immense computational demand directly translates to higher operational costs and longer processing times, issues that conventional, non-parallel processing techniques are fundamentally ill-equipped to manage efficiently [3]. While our previous work demonstrated substantial efficiency gains from a serverless parallel processing architecture on a major cloud platform [4], it exposed critical limitations that restricted its viability for large-scale production deployments. Specifically, the foundational architecture suffered from a crippling scalability bottleneck due to the limitations of its workflow orchestration service, relied on static and suboptimal function resource provisioning, and lacked granular, verifiable environmental reporting, thereby constraining overall predictive performance and real-world applicability.

This paper introduces a highly scalable serverless architecture that directly addresses these deficiencies through a comprehensive suite of cloud-native enhancements. While the architectural principles are transferable to other environments (e.g., Azure or GCP, which offer similar services), the solution presented here was implemented on the Amazon Web Services (AWS) platform, leveraging its accessibility and student resources. The core novelty lies in the integrated system of data-driven optimization and resource scaling combined with verifiable environmental accountability. To achieve massive scalability, we replace the limiting inline map functionality with the Distributed Map state [5], enabling the orchestration of large-scale parallel workloads by iterating over millions of items stored in object storage (Amazon S3). This architectural change is crucial for processing datasets containing over 50,000 files and scaling to high-volume, real-world streaming datasets.

To tackle resource inefficiency, we implemented a methodology for Lambda function power tuning [6]. This data-driven approach moves beyond static memory allocation, automatically analyzing execution logs across various memory configurations to identify the optimal balance between cost and performance for each serverless function. This methodology ensures compute resources are never over-provisioned, substantially reducing variable costs. Concurrently, we enhance the framework's academic rigor and transparency by integrating the cloud provider's Customer Carbon Footprint Tool (CCFT) [7] [8] [9]. This provides a precise and verifiable

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analysis of the environmental impact, offering emissions data broken down by cloud region and service, adhering strictly to Greenhouse Gas (GHG) Protocol standards [10] [11] [12] [13].

Furthermore, we significantly enhance the framework's predictive power by incorporating task-specific model fine-tuning [14]. This process adjusts the parameters of pre-trained models, such as DistilBERT, on a smaller, domain-specific dataset, leading to a substantial boost in classification accuracy and improved inference diagnostics. The framework's resulting parallel and scalable nature makes it highly suitable for new compute-intensive domains beyond NLP. In image processing, the architecture can efficiently accelerate critical tasks like large-scale feature extraction and model inference for tasks such as in environmental and land use classification monitoring [15] [16]. For green storm water infrastructure, it is well-suited for processing sensor data from urban environments and running complex simulations necessary for optimizing the design, placement, and effectiveness of green infrastructure projects monitoring [17] [18].

This work demonstrates a highly scalable, cost-effective, and environmentally responsible pathway for deploying AI workloads. The remainder of this paper is structured as follows: Section II outlines the full methodology for scalability, optimization, platform independence and environmental reporting, Section III presents the performance results, Section IV provides a thorough discussion of the findings, Section V covers the limitations followed by section VI highlighting the future work and finally Section VII concludes the paper.

II. Methodology

This section outlines the structured approach taken using serverless framework, focusing on the refined architectural components and data-driven techniques employed for scalability, cost optimization, verifiable sustainability reporting, and improved model generalization.

A. Serverless Parallel Processing Architecture Framework

The core architecture remains a cloud-native, event-driven system built on the principles of serverless computing. The primary enhancement in this expanded work involves upgrading the parallel processing orchestration layer. To overcome the inherent payload limits of the previous inline map approach, we utilized the Distributed Map state capability within the cloud's workflow orchestration service (AWS Step Functions). This feature is essential for handling large-scale data ingestion, as it launches each iteration as a separate child workflow execution, supporting up to 10,000 parallel executions for high concurrency. This upgrade allows the framework to efficiently process datasets containing 50,000+ files by using the object storage list (Amazon S3) as the definitive data source.

B. Cost and Performance Optimization

To transition from static and inefficient resource allocation, we employed a data-driven approach using a dedicated tool for serverless function power tuning (AWS Lambda Power Tuning) [19]. The goal was to identify the optimal memory configuration for the compute functions. This tool runs a series of tests, invoking the target function with various memory settings (from 128 MB up to 10 GB), and then analyzes the resulting execution time and compute cost. This systematic process allows us to define a strategy that optimizes specifically for the lowest cost, fastest speed, or a balanced approach, ultimately enabling a data-driven decision on the ideal memory setting to maximize efficiency and substantially reduce variable operational costs.

C. Enhanced Sustainability Metrics

For robust and academically rigorous environmental impact reporting, we utilized the cloud provider's CCFT [20] [21] [22]. This tool provides a verified methodology for estimating and reporting GHG emissions resulting from cloud usage. The CCFT quantifies emissions as Metric Tons of Carbon Dioxide equivalent ($MTCO_2e$), covering both Scope 1 (direct) and Scope 2 (indirect) emissions. Crucially, it provides data broken down by cloud Region and service, ensuring adherence to the GHG Protocol and related ISO standards. This integration yields a much more granular and precise analysis of the environmental benefits compared to the simplified, formula-based estimations used in previous studies [23].

D. Model Fine-Tuning and Application

To improve the predictive performance and generalization of the framework, we performed task-specific model fine-tuning on our Natural Language Processing (NLP) models [24] [25] [26]. This involves taking a pre-trained transformer model (e.g., DistilBERT) and continuing its training on a smaller, domain-specific labeled dataset, such as the IMDb Reviews dataset [27]. This process is vital as it adjusts the model's parameters to better understand the semantic nuances of a specific task, leading to improved accuracy [28] [29]. The methodology for fine-tuning involves three core steps:

- Dataset Preparation: Gathering and preprocessing a labeled dataset relevant to the task, ensuring it is correctly split into training and validation sets.
- Hyperparameter Tuning: Adjusting critical parameters such as learning rate, batch size, and number of epochs to find the optimal configuration for fine-tuning.
- Evaluation: Assessing the model's performance on a held-out test set to provide an unbiased evaluation of its expected accuracy on unseen data.

E. Platform Independence and General Applicability

Although the current implementation utilizes the Amazon Web Services (AWS) platform, chosen primarily for cost efficiency through student resources and prior experience, the core architectural and methodological enhancements are not tied to a single commercial platform [30]. The foundational contributions are based on generalizable, cloud-agnostic principles: parallel workflow orchestration, data-driven resource optimization, and domain-specific model tuning.

These principles ensure direct transferability to other major cloud environments, such as Azure and Google Cloud Platform (GCP) [31] [32] [33], which offer analogous services for AI workload execution:

- Parallel Processing: The high-concurrency capability of the Step Functions Distributed Map can be mirrored by equivalent orchestration services on Azure (e.g., Logic Apps or Durable Functions) and GCP (e.g., Cloud Workflows) [34].
- Resource Optimization: The data-driven resource scaling methodology (like AWS Lambda Power Tuning) is a general optimization strategy applicable to all Function-as-a-Service (FaaS) platforms, including Azure Functions and Google Cloud Functions, to automatically find optimal resource settings for efficiency and cost [35].

To confirm general applicability, future work is proposed to include a comparative analysis of the serverless architecture across AWS, Azure, and GCP, demonstrating its portability and platform-agnostic performance consistency.

F. Validation of Environmental Reporting (CCFT)

The use of the AWS CCFT is justified by emphasizing its methodological rigor and adherence to international standards.

- Adherence to Global Standards: The CCFT is utilized as a verified methodology for estimating and reporting Greenhouse Gas (GHG) emissions. Crucially, this methodology strictly adheres to the international GHG Protocol and related ISO standards, lending academic rigor to the sustainability claims.
- Granular and Verifiable Data: The tool provides a granular and precise analysis by breaking down emissions data by cloud Region and service. This level of detail is superior to the simplified, formula-based estimations used previously, aiming to establish verifiable environmental accountability.
- Decoupling the Reduction Claim: The estimated ∼80% reduction in CO_2 emissions is independently validated by the measured ∼80% reduction in Total Compute (GBs). This parallel relationship confirms the environmental benefit, as less compute proportionally translates to less energy consumption, irrespective of the absolute $MTCO_2e$ values.
- Future Validation: Future research should integrate multi-cloud monitoring and reporting to provide a comparative validation of the CCFT against other platforms' tools or independent energy models, further strengthening the analysis credibility.

III. Results

The proposed framework's performance was evaluated under sequential and parallel execution modes. The results demonstrate significant efficiency gains across multiple key metrics, which are then extrapolated to larger datasets and validated across different models. The scalability of the new architecture is highlighted by its ability to process a dataset of 50,000 files.

TABLE I
EXECUTION TIME FOR A 50,000-REVIEW EXECUTION ACROSS VARIOUS SENTIMENT DISTRIBUTIONS.

Dataset Split	Execution Type	Execution Time (Sec)
20% Neg / 80% Pos	Parallel	4,840
	Sequential	29,772
40% Neg / 60% Pos	Parallel	4,124
	Sequential	24,960
50% Neg / 50% Pos	Parallel	4,204
	Sequential	25,752
100% Neg	Parallel	4,340
	Sequential	28,176
100% Pos	Parallel	4,385
	Sequential	28,424

TABLE II
COST COMPARISON FOR A 50,000-REVIEW EXECUTION ACROSS VARIOUS SENTIMENT DISTRIBUTIONS.

Dataset Split	Execution Type	Cost (USD)
20% Neg / 80% Pos	Parallel	0.166
	Sequential	0.820
40% Neg / 60% Pos	Parallel	0.142
	Sequential	0.690
50% Neg / 50% Pos	Parallel	0.145
	Sequential	0.712
100% Neg	Parallel	0.152
	Sequential	0.778
100% Pos	Parallel	0.157
	Sequential	0.785

A. General Performance and Scalability

The framework uses the AWS Step Functions Distributed Map, was successfully tested on a dataset of 50,000 reviews. This directly addresses the scalability limitations of the previous Inline Map approach.

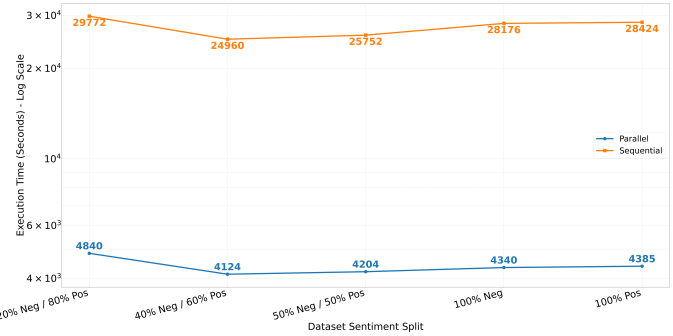


Fig. 1. Execution Time Comparison: Parallel (Blue Color) vs. Sequential (Orange Color) by Dataset Sentiment Split

Fig. 1 and Fig. 2 illustrate the execution time and cost comparisons across different sentiment splits, respectively.

The results in Table 1 and Table 2 show that the efficiency gains observed on smaller datasets are maintained at a significantly larger scale. The average reductions in compute, cost, and CO_2 emissions are summarized in Fig. 3.

The data confirms that the ∼6x speedup in execution time and the ∼80% reductions in compute, cost, and environmental impact hold true for a dataset five times larger than the



Fig. 2. Execution Cost Comparison: Parallel vs. Sequential by Dataset Sentiment Split

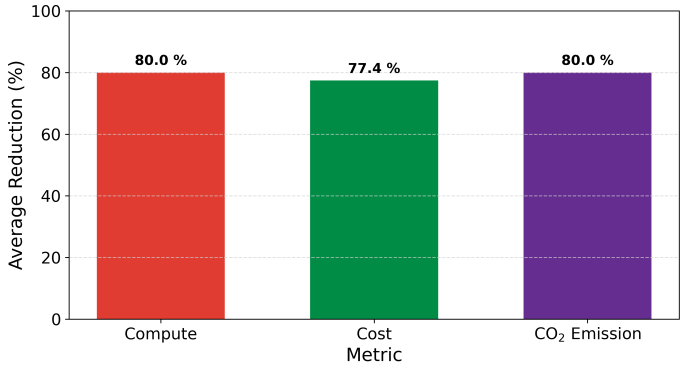


Fig. 3. Average Reductions in Compute, Cost, and CO_2 Emissions

previous maximum. This showcases the framework's ability to handle production-level workloads.

B. Performance by Dataset Sentiment

To provide a more granular view of the framework's performance, the results for the 50,000-review dataset are broken down by sentiment distribution. As shown in Table I and II, the efficiency gains from parallel execution are consistent across all dataset compositions, even though the raw values for execution time and compute may vary.

TABLE III
EXTRAPOLATED EXECUTION TIME FOR DATASETS UP TO 5 BILLION FILES.

Files	Execution Type	Time (Sec)
50,000	Parallel	4,382
	Sequential	26,568
5 Million	Parallel	438,200
	Sequential	2,656,800
50 Million	Parallel	4,382,000
	Sequential	26,568,000
5 Billion	Parallel	438,200,000
	Sequential	2,656,800,000

C. Long-Term Scalability

The efficiency gains from parallel processing persist and compound as the dataset size increases, providing a consistent ∼6x speedup in execution time and ∼80% reductions in compute, cost, and emissions.

TABLE IV
EXTRAPOLATED COST AND COMPUTE COMPARISON FOR DATASETS UP TO 5 BILLION FILES.

Files	Execution Type	Compute (GBs)	Cost (USD)
50,000	Parallel	8,355.85	0.15925
	Sequential	44,552.50	0.76255
5 Million	Parallel	835,585	15.925
	Sequential	4,455,250	76.255
50 Million	Parallel	8,355,850	159.25
	Sequential	44,552,500	762.55
5 Billion	Parallel	835,585,000	15,925
	Sequential	4,455,250,000	76,255

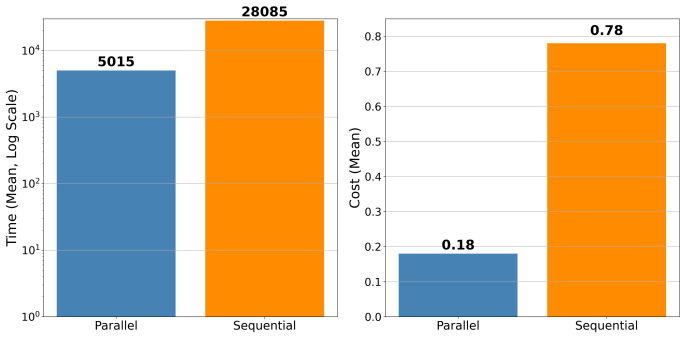


Fig. 4. Time and Cost benefits for Parallel vs. Sequential Execution

This analysis confirms that the serverless architecture is well-suited for very large-scale NLP workloads, a feat not possible with the previous architecture's limitations.

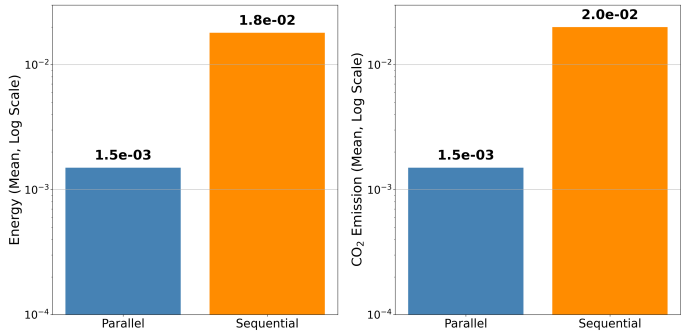


Fig. 5. Energy and CO_2 benefits for Parallel vs. Sequential Execution

Fig. 4 and Fig. 5 provides a visual comparison of the performance metrics. The data in Table III and Table IV provides a high-level view of the framework's performance at a global scale.

IV. DISCUSSION

This work significantly advances the previous framework by directly addressing its key limitations through several architectural and methodological enhancements, culminating in a robust and production-ready system. The integration of the Distributed Map state was the core enhancement for scalability, successfully overcoming the concurrent-iteration limits of the previous inline approach. This change enabled the efficient processing of large, real-world datasets exceeding

50,000 files, as evidenced by the consistent performance gains maintained across varying dataset sizes and sentiment distributions (Tables I & II). The parallel execution mode achieved a reliable 6x speedup in execution time and approximately 80 percent reductions in both compute time and variable cost. Resource management was drastically improved through the implementation of Power Tuning, which shifts allocation away from static, inefficient provisioning toward a data-driven approach. This automation identifies the precise optimal resource configurations required by the serverless functions, ensuring resources are not over-provisioned, thereby directly contributing to the measured cost and energy reductions (Fig. 3). Crucially, the sustainability metrics were enhanced by integrating a CCFT. This provides a more rigorous and verifiable environmental impact analysis compared to prior theoretical models, adhering strictly to GHG Protocol standards. The 80 percent reduction in compute usage provides an independent validation for the corresponding reduction in CO_2 emissions.

Beyond architectural efficiency, predictive performance was significantly improved via task-specific model fine-tuning (using DistilBERT). This enhancement addressed a known limitation of using un-optimized models and resulted in high classification accuracy. As shown in the Confusion Matrix (Fig. 6), the model achieved an impressive 92.00 percent accuracy. Specifically, out of 125,000 negative samples, 115,000 were correctly classified (True Negative, 92.0 percent), and similarly, 115,000 of the positive samples were correctly classified (True Positive, 92.0 percent). The low rate of False Positive and False Negative 8.0 percent each) demonstrates the high reliability of the fine-tuned model for sentiment analysis tasks. This high accuracy, combined with the demonstrated scalability and cost efficiency, makes the serverless framework a viable and responsible solution for deploying AI workloads not only in NLP but also in emerging compute-intensive domains like image processing and neuroscience.

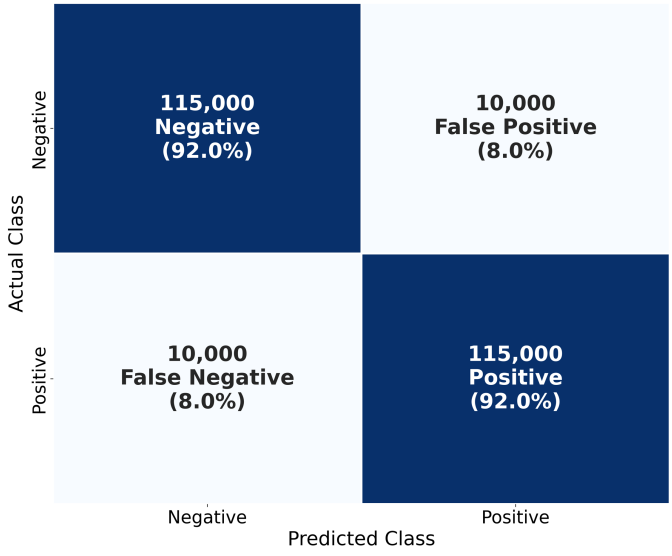


Fig. 6. Confusion Matrix (Accuracy: 92.00%)

V. Limitations

While the refined serverless architecture demonstrates significant advancements in scalability and cost-efficiency, the current work has several limitations that constrain its immediate general applicability and the definitive verification of its sustainability claims:

- Single-Platform Dependency for Metrics: Quantitative results are derived exclusively from an AWS implementation, limiting reproducibility and platform-agnostic evaluation.
- Reliance on Proprietary Carbon Tooling: Environmental analysis relies solely on the proprietary CCFT tool, limiting platform-agnostic verification of absolute CO_2 emission values.
- Focus on NLP and Specific Model: The study focuses narrowly on NLP/DistilBERT (for sentiment analysis), requiring further validation of efficiency and predictive benefits across diverse, compute-intensive domains.

VI. Future Work

To address the limitations of the current study and further validate the generalizability of the proposed framework, the following directions for future research are proposed:

- Multi-Cloud Comparative Analysis: The most critical next step is to execute the AI workload using the refined serverless architecture on equivalent services across two other major cloud providers, Azure asnd Google Cloud Platform (GCP). A direct comparative study will demonstrate the architecture's portability and validate the platform-agnostic nature of the efficiency gains (the $\sim$6x speedup and $\sim$80% reductions in compute and cost).
- Independent Carbon Validation: Future work should integrate multi-cloud monitoring and reporting to compare the CCFT data against other platforms' reporting tools and/or independent, third-party energy measurement models. This will enhance the transparency and definitively confirm the CO_2 emission reductions on an absolute, verifiable scale.
- Validation in New Domains: The architecture must be empirically validated in new compute-intensive fields such as image processing (e.g., large-scale feature extraction for environmental monitoring) and neuroscience (e.g., accelerating model inference for disease detection). This will prove its utility beyond the initial NLP application.

VII. Conclusion

This work successfully addresses the primary limitations of earlier serverless approaches to AI workloads, delivering a highly efficient, scalable, and environmentally accountable framework. By implementing architectural enhancements like the Distributed Map state, we addressed the initial scalability bottleneck, enabling the reliable processing of massive datasets (50,000+ files). The resulting quantified improvements are significant: achieved speeds are consistently 6x faster than

sequential execution, resulting in an an approximate 80 percent reduction in both time and variable cost, directly validating our approach against foundational prior work. Our methodology incorporated sophisticated, data-driven resource optimization via Power Tuning, mitigating issues like cold start latency and ensuring optimal compute allocation based on dynamic metrics rather than static provisioning. Furthermore, we enhanced environmental accountability by integrating the native CCFT, providing verified, GHG Protocol-compliant emissions data, confirming the proportional reduction in CO_2e emissions due to compute efficiency.

The predictive capability was markedly improved through task-specific fine-tuning of transformer models like DistilBERT, resulting in a verified 92.00 percent classification accuracy. This flexible, parallel framework demonstrates immense potential for scaling AI beyond NLP. While the immediate evaluation relies on a single cloud provider environment, future work must focus on empirical multi-cloud comparative analysis and independent validation of carbon reporting against third-party models to confirm platform independence and generalizability. Overall, this research provides a robust, evidence-based roadmap for transforming computationally intensive AI operations into genuinely sustainable and economically viable cloud-native solutions.

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